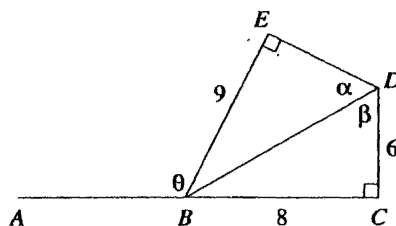


QUESTION ONE (Start a new answer booklet)

- (a) Express 225° in radians in terms of π .
- (b) An object is projected at an angle of 30° to the horizontal and with speed 40 m/s. What is its initial vertical speed?
- (c) Differentiate with respect to x :
- (i) $y = e^{4-x}$,
- (ii) $y = \sin^{-1} \sqrt{x}$.
- (d) Find primitives of:
- (i) $\frac{1}{2x}$,
- (ii) $\frac{1}{9+x^2}$.
- (e) Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{2x}$.
- (f) Let $P(x) = 2x^3 + x^2 - 7x - 6$.
- (i) Find the remainder when $P(x)$ is divided by $(x+2)$.
- (ii) Suppose $P(x)$ has roots α , β and γ . Find the value of $\alpha\beta\gamma$ without finding the roots.
- (g) A polynomial of degree 5 is divided by another polynomial of degree 2. What is the maximum degree of the remainder?

QUESTION TWO (Start a new answer booklet)

- (a) Given $t = \tan \frac{\theta}{2}$, write down an expression in terms of t for $\cos \theta$.
- (b) The polynomial $f(x) = x^3 + ax^2 + b$ is exactly divisible by $(x - 2)$, and has a remainder of 5 when divided by $(x - 1)$. Find the values of a and b .
- (c) The area under $y = \frac{5}{\sqrt{25 + x^2}}$ between $x = 0$ and $x = 5$ is rotated about the x -axis.
- (i) Write down the integral for the volume of this solid.
- (ii) Hence find the volume.
- (d)

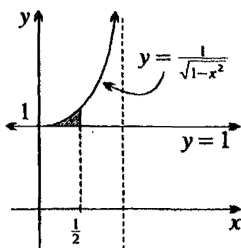


In the diagram above, $BC = 8$, $CD = 6$ and $BE = 9$, and the angles at C and E are right angles as shown.

- (i) Find the lengths of BD and ED .
- (ii) Show that $\theta = \alpha + \beta$.
- (iii) Hence or otherwise, show that $\sin \theta = \frac{27 + 4\sqrt{19}}{50}$.

QUESTION THREE (Start a new answer booklet)

(a)



- (i) Write down the area of the shaded region in terms of an integral.
 - (ii) Hence show that this area is $\frac{\pi - 3}{6}$.
- (b) The velocity of a particle is given by $v = t^2 - 2t$, and it starts at $x = 3$.
- (i) When is the particle moving to the left?
 - (ii) Find the displacement x as a function of time.
 - (iii) What is the closest the particle gets to the origin?
- (c) The monic polynomial $g(x)$ of degree 6 has a zero at $x = -3$, a double zero at $x = -1$, and a triple zero at $x = 1$.
- (i) Write down $g(x)$ in factored form.
 - (ii) Find the y -intercept of the graph of $g(x)$.
 - (iii) Sketch $y = g(x)$, without using calculus, clearly showing the behaviour at each x -intercept.

QUESTION FOUR (Start a new answer booklet)

- (a) Write $\cos 2t + \sin 2t$ in the form $A \cos(2t - \epsilon)$.
- (b) (i) Sketch $y = e^x$ and $y = 4 - x^2$ on the same number plane, showing all intercepts with the axes.
- (ii) Deduce from your answer to part (i) the number of solutions to the equation $e^x = 4 - x^2$.
- (iii) Let $g(x) = e^x + x^2 - 4$. Given the initial estimate for a zero of this function is $x_0 = 1$, apply Newton's method once to find a better approximation x_1 . Give your answer to three significant figures.

QUESTION FIVE (Start a new answer booklet)

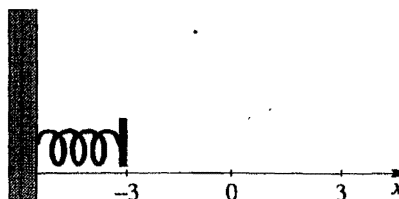
- (a) When an object has displacement x , its acceleration is given by

$$\ddot{x} = -3(2 - x)^2.$$

As the object passes through the origin, it has a velocity of 4 m/s.

- (i) Show that $v^2 = 2(2 - x)^3$.
- (ii) Explain why the object can never be to the right of $x = 2$.
- (iii) Show that the object's velocity can never be negative.

(b)



A plate attached to a spring mounted on a wall has mean position $x = 0$. The spring is compressed so that the plate is at $x = -3$, as shown above. The plate is then released from rest, and its displacement x metres at time t seconds satisfies

$$\ddot{x} = -\frac{16}{9}x.$$

- (i) Given $x = -A \cos nt$, find the values of A and n .
- (ii) What is the period of this motion?
- (iii) What is the speed of the plate as it passes through the origin?
- (iv) The experiment is repeated with a light ball placed in front of the plate. Explain why the ball separates from the plate as it passes through the origin.

QUESTION SIX (Start a new answer booklet)

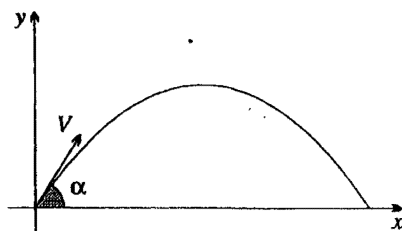
- (a) Consider the cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$, which has roots α , β and γ such that $\gamma = \alpha + \beta$.

(i) Show $\alpha + \beta = -\frac{b}{2a}$.

(ii) Also show $\alpha\beta = \frac{2d}{b}$.

(iii) Hence show that α and β are also roots of $x^2 + \frac{b}{2a}x + \frac{2d}{b} = 0$.

(b)



A ball is thrown on level ground at an initial speed of V m/s and at an angle of projection α as in the diagram above. You may assume that t seconds after release, the horizontal and vertical displacements are:

$$x = Vt \cos \alpha$$

$$y = Vt \sin \alpha - \frac{1}{2}gt^2.$$

- (i) Find the Cartesian equation of the trajectory.

(ii) Hence show that the range is $\frac{V^2 \sin 2\alpha}{g}$.

- (iii) When the ball is thrown with speed $V = 30$ m/s, it lands 45 metres away.

(α) Given $g = 10$ m/s², find α .

- (β) A 2 metre high fence is placed 40 metres away from the thrower. Examine each trajectory to see whether it is still possible.

SGS.

VI 3 Unit Mid Yr. 1999

Solutions

1/ (a) $\frac{5\pi}{4}$

(b) $y = 40 \sin 30^\circ$
 $= 20 \text{ m/s.}$

(c) (i) $y = e^{4-x}$
 $y' = -e^{4-x}$

(ii) $y = \sin^{-1} \sqrt{x}$
 $y' = \frac{1}{2\sqrt{x}} \cdot \frac{1}{\sqrt{1-x}}$
 $= \frac{1}{2\sqrt{x-x^2}}$

(d) (i) $y' = \frac{1}{2x}$
 $y = \frac{1}{2} \log x \quad (+c)$

(ii) $y' = \frac{1}{9+x^2}$
 $y = \frac{1}{3} \tan^{-1} \frac{x}{3} \quad (+c)$

(e) $\lim_{x \rightarrow 0} \frac{\sin x}{2x} = \frac{1}{2} \lim_{x \rightarrow 0} \frac{\sin x}{x}$
 $= \frac{1}{2}$

(f) (i) $P(-2) = 2 \times (-8) + 4 - 7 \times (-2) - 6$
 $= -4$

(ii) $\alpha\beta\gamma = -\frac{d}{a} = 3$

g) degree 1.

2/ (a) $\cos \theta = \frac{1-t^2}{1+t^2}$

(b) $f(4) = 8 + 4a + b = 0 \quad (1)$

$f(1) = 1 + a + b = 5 \quad (2)$

(1) - (2) yields

$7 + 3a = -5$

$a = -4$

so $b = 8$

and $f(x) = x^2 - 4x^2 + 8.$

(c) (i) $v = \pi \int_0^5 \frac{25}{25+x^2} dx$

(ii) so $v = 25\pi \left[\frac{1}{5} \tan^{-1} \left(\frac{x}{5} \right) \right]_0^5$
 $= 5\pi [\tan^{-1} 1 - 0]$
 $= \frac{5\pi^2}{4} \text{ cubic units.}$

(d) (i) $BD = 10$ (by Pyth.)

$ED = \sqrt{19}$ (by Pyth.)

(ii) $\angle AED = 90 - \alpha$ ($\angle$ sum of $\triangle AED$)

$\angle CED = 90 - \beta$ ($\angle$ sum of $\triangle CED$)

so $\angle BEC = 180^\circ - (\alpha + \beta)$

$= 180 - \theta$

thus $\theta = \alpha + \beta.$

[a use the fact that $\triangle BEC$ is a cyclic quad.]

(iii) $\sin \theta = \sin(\alpha + \beta)$

$= \sin \alpha \cos \beta + \cos \alpha \sin \beta$

$= \frac{9}{10} \cdot \frac{6}{10} + \frac{\sqrt{19}}{10} \cdot \frac{8}{10}$

$= \frac{27 + 4\sqrt{19}}{50}$

3/ (a) (i) $A = \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{1-x^2}} - 1 dx$

(ii) $A = [\sin^{-1} x - x]_0^{\frac{1}{2}}$

$= \frac{\pi}{6} - \frac{1}{2}$

$= \frac{\pi-3}{6}$

(b) (i) ie $v < 0$ so $t^2 - 2t < 0$

thus $0 < t < 2$

(ii) $x = \int v dt$

$= \frac{t^3}{3} - t^2 + c$

at $t=0, x=c=3$

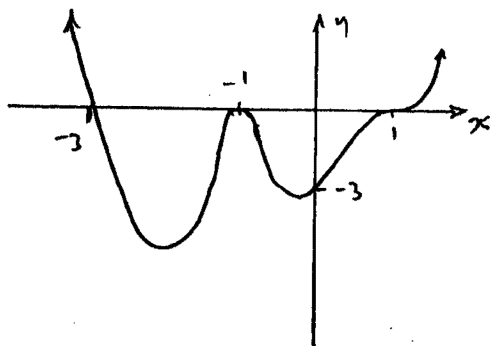
thus $x = \frac{t^3}{3} - t^2 + 3$

(iii) at $t=2, x = \frac{5}{3}.$

$$g(x) = (x+3)(x+1)^2(x-1)^2$$

(ii) $g(0) = -3$

(iii)



4/ (a) $\cos 2t + \sin 2t = A \cos(2t - \epsilon)$
 $= A \cos 2t \cos \epsilon + A \sin 2t \sin \epsilon$

equating coeffs. of $\sin 2t$ and $\cos 2t$:

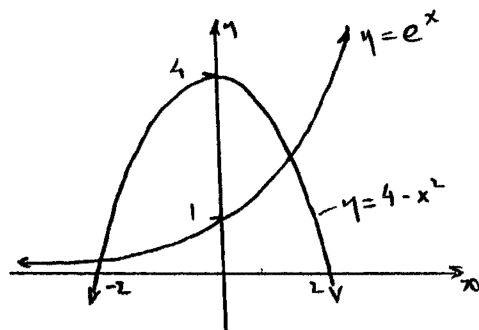
$$A \cos \epsilon = 1$$

$$A \sin \epsilon = 1$$

hence $A = \sqrt{2}$ and $\epsilon = \pi/4$ since it's in 1st quad.

so $\cos 2t + \sin 2t = \sqrt{2} \cos(2t - \frac{\pi}{4})$

(b)(i)



(ii) They intersect twice

(iii) $g(x) = e^x + x^2 - 4$ so $g(1) \doteq 0.2817$

$g'(x) = e^x + 2x$ so $g'(1) \doteq 4.7182$

$$x_1 = 1 - \frac{g(1)}{g'(1)}$$

$$\doteq 1.0597...$$

$$\doteq 1.06 \text{ to 3 sig fig.}$$

5(a)(i) $\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = -3(2-x)^2$

so $\frac{1}{2} v^2 = (2-x)^3 + \frac{c}{2}$ for some constant c .

at $x = 0$, $v = 4$ so $c = 0$

and $v^2 = 2(2-x)^3$

(ii) require $v^2 \geq 0$ so $2(2-x)^3 \geq 0$

hence $x \leq 2$.

(iii) for v to become $-ve$, it must first be zero.

when $v = 0$, $x = 2$

when $x = 2$, $\ddot{x} = 0$

Thus the particle is at rest and remains so [since $\ddot{x}$ is strictly a function of x]. Hence v is never $-ve$.

or

let $v = \sqrt{2} (2-x)^{3/2}$ which is true at least once. Then

$$t = \frac{\sqrt{2}}{\sqrt{2-x}} + k \text{ by integration.}$$

From this we see x never reaches 2. so $v \neq 0$ and hence v never changes sign.

(b) (i) $r^2 = \frac{16}{9}$ so $r = \frac{4}{3}$

at $t = 0$ $x = -A = -3$

so $A = 3$

and $x = -3 \cos\left(\frac{4t}{3}\right)$

(ii) period = $\frac{2\pi}{(4/3)} = \frac{3\pi}{2}$

(iii) $\dot{x} = 4 \sin\left(\frac{4t}{3}\right)$

so at the origin speed = 4 m/s .

(iv) The plate begins to decelerate but the ball keeps going at the same rate.

[It may be tempting for 4u boys to introduce arguments about friction and the like. Given the information in the question, the above still stands - i.e. no friction etc....]

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\text{so } 2(\alpha + \beta) = -\frac{b}{a}$$

$$\text{and } \alpha + \beta = -\frac{b}{2a} \quad (1)$$

$$(ii) \quad \alpha\beta\gamma = -\frac{d}{a}$$

$$\text{so } \alpha\beta(\alpha + \beta) = -\frac{d}{a}$$

$$\alpha\beta\left(-\frac{b}{2a}\right) = -\frac{d}{a}$$

$$\text{hence } \alpha\beta = \frac{2d}{b} \quad (2)$$

(iii) For the given equation, the sum of the roots is $-\frac{b}{2a}$ and $\alpha + \beta = -\frac{b}{2a}$; also the product of the roots is $\frac{2d}{b}$ and $\alpha\beta = \frac{2d}{b}$. Hence α and β are the roots.

or
Solve equations (1) and (2) simultaneously.

$$(b)(i) \quad t = \frac{x}{v \cos \alpha}$$

$$\text{so } y = v \sin \alpha \cdot \frac{x}{v \cos \alpha} - \frac{g}{2} \left(\frac{x}{v \cos \alpha} \right)^2$$

$$= x \tan \alpha - \frac{g}{2v^2} x^2 \sec^2 \alpha$$

$$(ii) \quad y = x \sec^2 \alpha \left[\sin \alpha \cos \alpha - \frac{g}{2v^2} x \right]$$

so the range is given by

$$\frac{gx}{2v^2} = \sin \alpha \cos \alpha$$

$$\text{or } x = \frac{v^2 \sin 2\alpha}{g}$$

$$(iii)(a) \quad 45 = \frac{900 \sin 2\alpha}{10}$$

$$\text{so } \sin 2\alpha = \frac{1}{2}$$

$$\text{thus } \alpha = 15^\circ \text{ or } 75^\circ$$

$$(b) \quad \text{at } x = 40$$

$$\text{if } \alpha = 15^\circ$$

$$y \doteq 1.19 < 2 \text{ m}$$

$$\text{if } \alpha = 75^\circ$$

$$y \doteq 16.59 > 2 \text{ m}$$

thus $\alpha = 15^\circ$ is not possible and hence $\alpha = 75^\circ$.

$$7(a)(i) \text{ in the } \Delta \quad C = \pi - (A+B)$$

so the sine rule with angle C gives

$$\frac{\sin [\pi - (A+B)]}{c} = \frac{\sin B}{b}$$

$$\text{but } \sin (\pi - \theta) = \sin \theta$$

$$\text{hence } \frac{\sin (A+B)}{c} = \frac{\sin B}{b}$$

(ii) expand LHS to get

$$\frac{\sin A \cos B + \cos A \sin B}{c} = \frac{\sin B}{b}$$

$$\times \frac{cb}{\cos B} \text{ to get}$$

$$b \sin A + b \cos A \tan B = c \tan B$$

$$\text{so } \tan B (c - b \cos A) = b \sin A$$

$$\tan B = \frac{b \sin A}{c - b \cos A}$$

$$= \frac{\sin A}{\frac{c}{b} - \cos A}$$

(iii) The angle A is the included angle between sides b and c, and hence ΔABC is unique by the SAS congruence test.

$$(b)(i) \text{ in } \Delta ABD \quad a^2 = b^2 + 1 - 2b \cos \theta \quad (1)$$

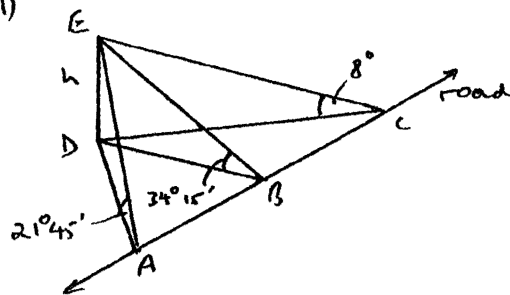
$$\text{in } \Delta CBD \quad c^2 = b^2 + 1 - 2b \cos (\pi - \theta)$$

$$\text{or } c^2 = b^2 + 1 + 2b \cos \theta \quad (2)$$

(1) + (2) yields

$$a^2 + c^2 = 2b^2 + 2$$

(ii)



$$DC = h \cot 8^\circ = h\gamma$$

$$DB = h \cot 34^\circ 15' = h\beta$$

$$DA = h \cot 21^\circ 45' = h\alpha$$

--- cd

part (i)

$$(h\alpha)^2 + (h\gamma)^2 = 2(h\beta)^2 + 2$$

rearranging

$$h^2(\alpha^2 + \gamma^2 - 2\beta^2) = 2$$

$$h = \sqrt{\frac{2}{\alpha^2 + \gamma^2 - 2\beta^2}}$$

$$\doteq 0.195 \text{ km}$$

$$= 195 \text{ m to nearest metre.}$$

$$\text{or } EC = h \sec 8^\circ = h\gamma$$

$$EB = h \sec 34^\circ 15' = h\beta$$

$$EA = h \sec 21^\circ 45' = h\alpha$$

and then proceed as above.

(c) Let the sides be

$$b-d, b, b+d$$

then by (b) (i)

$$(b-d)^2 + (b+d)^2 = 2b^2 + 2$$

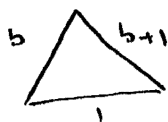
$$\text{so } 2b^2 + 2d^2 = 2b^2 + 2$$

$$\text{so } d^2 = 1$$

$$\text{and } d = 1 \quad (\text{the -ve soln is redundant})$$

The sides are then $b-1, b, b+1$

so $\triangle BCD$ has dimensions



which is not possible for a $\triangle$.

or

$$\text{Let } b = \frac{a+c}{2}$$

$$\text{so } a^2 + c^2 = 2\left(\frac{a+c}{2}\right)^2 + 2$$

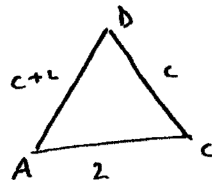
$$2a^2 + 2c^2 = (a+c)^2 + 4$$

$$\text{thus } (a-c)^2 = 4$$

$$\text{and } a-c = 2 \quad (\text{the -ve soln is redundant})$$

$$\text{so } a = c+2$$

$\triangle ABC$ then has the dimensions



which is not possible for a $\triangle$.

[There are other variations on the same basic proof.]

D.J.W.

20/5/99.